

A mechanistic model simulating ascospore infections by *Erysiphe necator*, the powdery mildew fungus of grapevine

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A new dynamic model for *Erysiphe necator* ascospore infections on grapevine was developed. Between budbreak of vines and the time when the pool of ascospores is depleted, the model uses weather data for calculating, at daily intervals: curve of ascospore maturation; ascospore discharge events and relative proportion of the discharged ascospores; infection periods and their relative infection severity; and progress of latency period and time when secondary infections should begin. The model was validated over a 4-year period (2005–2008) in 26 vineyards in Italy by comparing model predictions with actual observations of the first seasonal symptoms of powdery mildew. The model showed high sensitivity, specificity and accuracy. Proportions of true and false positive predictions were TPP = 0.94 and FPP = 0.26, respectively. Because a proportion of predicted infection periods did not result in actual disease onset, confidence was higher for prediction of non-infections than for prediction of infections. Most of the false positive predictions occurred in the earlier growth stages of the host, when the surface area of susceptible tissue may be very small so that the probability that ejected ascospores land on susceptible tissue is low. An equation was then developed to describe the probability that a predicted infection period results in disease onset as a function of the growth stage of vines at the time of prediction. The new model should improve early season powdery mildew management by helping vineyard managers schedule fungicide sprays or schedule the scouting of the vineyard for detection of first disease signs.

Keywords: disease modelling, disease prediction, environmental conditions, primary inoculum, *Vitis vinifera*

Introduction

Erysiphe necator (formerly *Uncinula necator*) is the causal agent of powdery mildew, a major grapevine disease throughout the world. The fungus can infect all green tissues of the grapevine but causes the most damage to leaves and berries. On leaves, it appears as a white or greyish-white powder covering the upper and lower surfaces; heavily infected leaves may become dull, desiccate and drop prematurely in the autumn. Fruit infection may appear white and powdery or dark and dusty, and may sometimes result in shrivelling or cracking of the berries.

Powdery mildew often causes substantial reductions in crop yield and wine quality. The disease is typically controlled by periodic applications of foliar fungicides, with four to eight fungicide sprays applied each growing season. Even though fungicide schedules often recommend that sprays begin at fixed growth stages, in many cases sprays are not applied until the disease becomes evident in the vineyard (Magarey *et al.*, 2000). This approach is not always successful. The disease does not become

evident until several weeks after it has begun to spread, and at that time the epidemic is so well advanced that control is difficult or impossible, even with repeated applications of systemic fungicides (Bertocchi *et al.*, 2010; Sozzani *et al.*, 2010). These late fungicides are ineffective because fungicides must begin promptly to control primary ascospore infection and be repeated at appropriate intervals to control secondary conidial infection. Monitoring protocols (Emmett *et al.*, 1998; Magarey *et al.*, 2000) and weather-driven models (Chellemi & Marois, 1991; Kast, 1997; Gubler *et al.*, 1999; Carisse *et al.*, 2009) have been developed to determine necessary spray intervals, but timing of the first treatment of the season is mostly dependent on the fungicide used and growth stage of vines (e.g. Magarey *et al.*, 2000; Hartman & Beale, 2008; Pscheidt & Ocamb, 2009).

Limited research has been done concerning the timing of the first application of fungicides against powdery mildew relative to epidemiological parameters. Kast (1997) developed a model based on data collected for more than 50 years in the wine regions of Württemberg and Rheinhessen in Germany. In this model, the first spray date is determined using a rough indexing of the disease severity for the vine site in the preceding year and the mean of the lowest temperatures in the two

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previous winters. Based on this information and the date when the 'three leaves unfolded' development stage occurs in the current season, a time lag is calculated. In another approach, Carisse *et al.* (2009) suggested that managers should scout the vineyard for the first disease signs once suitable environmental conditions had occurred (≥ 2.5 mm of rainfall over a ≥ 4 -h period at 6–24°C), and apply the first spray when the disease had appeared. These models are both empirical and are more focused on timing fungicide applications than on modelling the influence of weather conditions on the pathogen's behaviour during the primary infection cycle.

In this work, a new weather-driven, dynamic model was constructed and evaluated to simulate the ecological behaviour of the *E. necator* ascocarp population in spring and the processes involved in the primary disease cycle. Such a model is a first step in developing a tool for a successful, season-long powdery mildew control based on reducing early season inoculum and subsequent infection (Gubler *et al.*, 2006).

Materials and methods

Primary disease cycle processes

Erysiphe necator is a heterothallic fungus (Smith, 1970), i.e. formation of ascocarps (chasmothecia, formerly cleistothecia; Braun *et al.*, 2002) begins in mid- to late summer, when two opposite mating types are in close proximity on the surface of infected tissue and the antheridium and oogonium combine (Schnathorst, 1965; Gadoury & Pearson, 1988). Once initiated, the chasmothecia mature in four stages: initially they are white, then yellow, then brown, and finally they become black at maturity. During maturation, chasmothecia are fixed to the powdery mildew colony by anchorage hyphae; when chasmothecia are morphologically mature, characteristic appendages appear, the anchorage breaks down and the fruiting bodies are dispersed by rain splashes (Gadoury & Pearson, 1988; Cortesi *et al.*, 1995; Jailloux *et al.*, 1998; Füzi, 1999). The rains deposit the chasmothecia onto the bark of the vine trunk, onto the soil, or onto leaf litter on the soil surface; however, survival of these ascocarps is consistently higher in the cracks of the trunk than on the other substrates (Gadoury & Pearson, 1988). Chasmothecia swell with water and forcibly discharge ascospores into the air; ascospore discharge occurs repeatedly between budbreak and bloom of the grapevine or shortly thereafter (Pearson & Gadoury, 1987; Rossi *et al.*, 2010). The ascospores are blown to nearby leaves (usually near the vine where chasmothecia had overwintered) where they germinate, form appressoria, and start the growth of a primary fungal colony on the surface of the infected tissue (Gadoury & Pearson, 1990a). After colonies from ascosporic infections have developed, the fungus can asexually propagate via large numbers of conidia that disperse (Carisse *et al.*, 2009) and re-infect additional leaves; the leaf

epidemic provides a source of inoculum for the infection of developing berries (Calonnec *et al.*, 2006).

Modelling approach

The model was elaborated according to principles of 'systems analysis' (Leffelaar & Ferrari, 1989). Previously described processes were separated into different stages, and changes from one stage to another depend on environmental conditions, as shown in the relational diagram of the model (Fig. 1 and Table 1). Stages of the primary disease cycle are considered as state variables, and rate variables or switches regulate changes in such variables. Influencing weather conditions are considered as parameters or intermediate variables.

Model description

Hourly meteorological data of temperature (T, °C); relative humidity (RH, %); rainfall (R, mm); and duration of wetness (WD, h) were used as inputs for the model.

The first state variable consists of the ascocarp population at complete leaf fall, which will overwinter and produce mature ascospores during the next grape-growing season (OCH, overwintered chasmothecia). Because chasmothecia are produced from summer to autumn and are progressively dispersed to the grapevine bark until

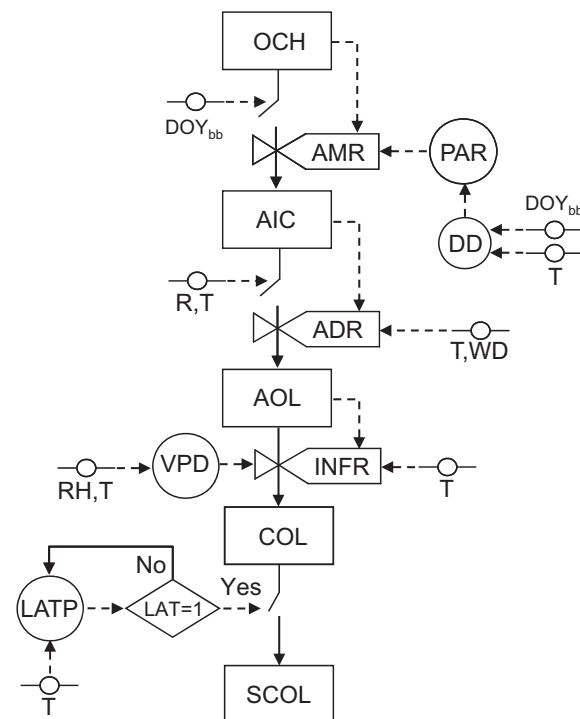


Figure 1 Relational diagram of the model simulating the ascosporic infection cycle of *Erysiphe necator* on grapevine. Legend: □ state variable; → flux and direction of states; → flux and direction of information; —○— parameter (i.e. input variable); ○ intermediate variable; ◇ switch; ⋈ valve in a flux (rate) (see Table 1 for acronym key).

Table 1 List of variables used in the model of ascospore *Erysiphe necator* infection on grapevine

Acronym	Description	Unit
OCH	Overwintered chasmothecia	Number (set at 1)
AIC	Ascospores in chasmothecia	Number (0 to 1)
AOL	Ascospores on grape leaves	Number (0 to 1)
COL	Colony-forming ascospores on grape leaves	Number (0 to 1)
SCOL	Sporulating colonies on grape leaves	Number (0 to 1)
AMR	Ascospore maturation rate	Number (0 to 1)
ADR	Ascospore discharge rate	Number (0 to 1)
INFR	Infection rate	Number (0 to 1)
PAR	Proportion of ascospores ready for discharge	Number (0 to 1)
LAT	Latent period, days elapsed from infection to sporulation	Number of days
LATP	Daily progress of latency	Number (0 to 1)
DOY _{bb}	Day of the year at budbreak of vines	Number
DD	Degree-days (base 10°C)	Number
T	Air temperature	°C
Teq	Temperature equivalent	Number (0 to 1)
Tmin	Minimum temperature for infection to occur	°C
Tmax	Maximum temperature for infection to occur	°C
R	Rainfall	mm
RH	Relative humidity	%
VPD	Vapour pressure deficit	hPa
WD	Wetness duration	h

leaf fall is complete (Rossi *et al.*, 2010), the model assumes that the vineyard's population of chasmothecia is defined on the 1 January of the following year and sets OCH = 1.

Starting from the day of budbreak (DOY_{bb}, i.e. the day of the year when plants have leaf tip visible), the model calculates the cumulative proportion of mature and ready-for-discharge ascospores (PAR, proportion of ascospores ready for discharge) as a function of degree-days (DD, base 10°C) accumulated before and after DOY_{bb}, using Eqn 1 from Rossi *et al.* (2010):

$$\begin{aligned} \text{PAR}_i &= \exp[-\alpha \cdot \exp(-\beta \cdot \text{DD}_i/100)] \\ &\text{with } \alpha = 1.95 \pm 0.188; \beta = 1.91 \pm 0.293 \quad (1) \\ \text{DD}_i &= \sum_{\text{DOY}_{bb}}^i (T_i - 10); \\ &\text{if } (T_i - 10) < 0 \text{ then } (T_i - 10) = 0 \end{aligned}$$

The model then calculates a daily rate of ascospore maturation (AMR, ascospore maturation rate) that regulates the flow from OCH to the second state variable, i.e. mature and ready-for-discharge ascospores in the fruiting bodies (AIC, ascospores in chasmothecia). AMR is the first derivative of PAR, and therefore:

$$\text{AIC}_i = \sum_{\text{DOY}_{bb}}^i \text{AMR}_i \cdot \text{OCH} \quad (2)$$

with

$$\text{AMR}_i = f'(\text{PAR}_i)$$

where $f'(\text{PAR})$ is the first derivative of PAR.

Ascospores in the AIC stage are discharged onto leaves (AOL, ascospores on grape leaves) whenever there is at least 2.5 mm of rainfall (Gadoury & Pearson, 1990b; Jailloux *et al.*, 1999; Rossi *et al.*, 2010) and T is not limiting, at a rate (ADR, ascospore discharge rate) that depends on T and wetness duration (WD, hours of the day with wetness). Ascospores that go from AIC to AOL stage in any discharge event occurring on day i are calculated as follows:

$$\text{AOL}_i = \text{AIC}_i \cdot \text{ADR}_i \quad (3)$$

when $R_i < 2$ mm, or $T < 4^\circ\text{C}$, or $T > 30^\circ\text{C}$,

$$\text{ADR}_i = 0 \quad (4)$$

when $R_i \geq 2$ mm and $4 \leq T \leq 30^\circ\text{C}$,

$$\text{ADR}_i = 1 - \delta \cdot \exp(-\lambda \cdot T_i^2 \cdot \text{WD}_i) \quad (5)$$

with: $\delta = 0.969 \pm 0.024$; $\lambda = 0.0004 \pm 0.00003$

Equation 5 was elaborated using the data from Gadoury & Pearson (1990b): the percentages of the chasmothecia releasing ascospores at different T and WD were expressed as proportions (i.e. on a scale of 0–1) and fitted to the above-mentioned equation (Fig. 2a); goodness-of-fit of this equation was high because the adjusted coefficient of determination (R_{adj}^2) was high (0.97), the standard error of model estimates was low (reported in Eqn 5), the root mean standard error (a measure of the differences between values predicted by the model and the values actually observed; Nash & Sutcliffe, 1970) was low (0.008), and the distribution of residuals did not show systematic deviations (not shown).

Once in the AOL stage, ascospores germinate, form appressoria and cause infection at a rate (INFR, infection rate) that depends on T and vapour pressure deficit (VPD, in hPa). INFR determines the proportion of colony-forming ascospores on leaves (COL). Ascospores that go from AOL to COL stage on any day i after a discharge event are calculated as follows:

$$\text{COL}_i = \text{AOL}_i \cdot \text{INFR}_i \quad (6)$$

when $T_i < 5^\circ\text{C}$ or $T_i > 31^\circ\text{C}$,

$$\text{INFR}_i = 0 \quad (7)$$

when $5 \leq T_i \leq 31^\circ\text{C}$,

$$\text{INFR}_i = \{ [\varphi \cdot \text{Teq}_i^v \cdot (1 - \text{Teq}_i)]^\zeta \} \cdot \exp(-v \cdot \text{VPD}_i) \quad (8)$$

with: $\varphi = 7.391 \pm 0.264$; $v = 2.403 \pm 0.105$; $\zeta = 0.892 \pm 0.078$; $v = 0.221 \pm 0.017$

$$\text{Teq}_i = (T_i - \text{Tmin}) / (\text{Tmax} - \text{Tmin}); \text{Tmin} = 5^\circ\text{C};$$

$$\text{Tmax} = 31^\circ\text{C}$$

$$\text{VPD}_i = (1 - \text{RH}_i/100) \cdot 6.11 \cdot \exp[(17.47 \cdot T_i) / (239 + T_i)]$$

(from Buck, 1981)

Like Eqn 5, Eqn 8 was elaborated using the data from Gadoury & Pearson (1990b): the percentage of the ascospores that germinate and form appressoria at different

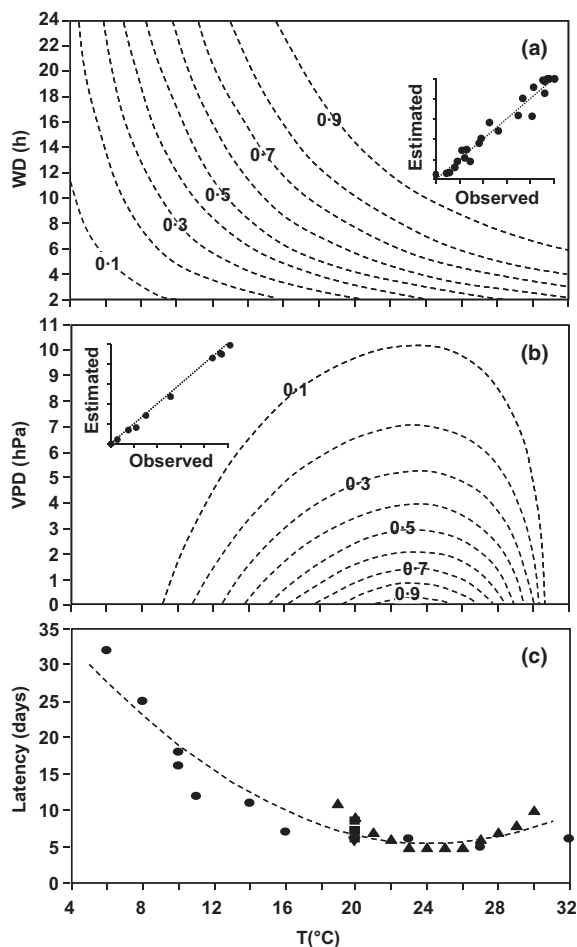


Figure 2 Relationship between environmental conditions and ascospore discharge rate (a), ascospore infection rate (b), and length of latency period (c) according to model Eqns 5, 8 and 10, respectively, for ascosporic *Erysiphe necator* infection on grapevine. T is air temperature, WD is wetness duration, VPD is vapour pressure deficit. The inset graphs in (a) and (b) show the correspondence between data from Gadoury & Pearson (1990b) and the data estimated by Eqns 5 and 8, respectively (dotted line is the line of perfect agreement between observed and estimated). In (c) points are data from Delp (1954) (●), Chellemi & Marois (1991) (▲) and Gadoury & Pearson (1988) (■).

T and RH were expressed as proportions (i.e. on a 0–1 scale) and fitted to Eqn 8 (Fig. 2b). For this equation, goodness-of-fit was high for both T ($R_{adj}^2 = 0.99$; root mean square error (RMSE) = 0.0015) and VPD ($R_{adj}^2 = 0.99$; RMSE = 0.023), standard error of the estimated parameters was low (see Eqn 8), and the residual distribution did not show systematic deviations (not shown).

After a latency period (LAT), which depends on T, the powdery mildew colony begins to sporulate (SCOL); the fresh conidia produced on these colonies will start the secondary infection cycles. The daily progress of latency (LATP) is calculated as follows:

$$LATP_i = 1/LAT_i \quad (9)$$

with:

$$LAT_i = \gamma + \psi \cdot T_i - \tau \cdot T_i^2 \quad (10)$$

with: $\gamma = 44.7 \pm 1.95$; $\psi = 0.067 \pm 0.006$; $\tau = 3.244 \pm 0.219$

When the accumulation of $LATP_i$ is ≥ 1 , colonies enter into the SCOL stage. Equation 10 was calculated by fitting data from Delp (1954), Pearson & Gadoury (1987), Gadoury & Pearson (1988) and Chellemi & Marois (1991), to a quadratic equation with $R_{adj}^2 = 0.89$ and normalized RMSE = 0.0095 (Fig. 2c).

Model output

During the period between budbreak of the vines and the time when the pool of ascospores is depleted (i.e. PAR = 1), the model produces the following output at daily intervals: (i) curve of ascospore maturation (i.e. PAR) (Fig. 3b); (ii) occurrence of ascospore discharge events (Fig. 3b); (iii) proportion of the seasonal ascospores that are discharged in each discharge event (i.e. AIC at that time) (Fig. 3b); (iv) occurrence of infection periods (i.e. when COL > 0) (Fig. 3c); (v) proportion of the

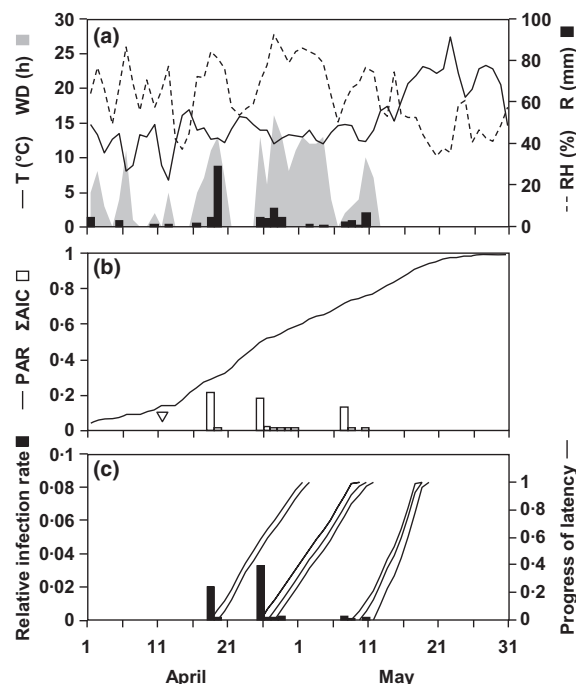


Figure 3 (a) Example of model input (air temperature, T; wetness duration, WD; relative humidity, RH; and rainfall, R); (b) and (c) correspondent model output: dynamics of ascospore maturation (PAR, proportion of ascospores ready for discharge); proportion of seasonal ascospores discharged (ΣAIC , ascospores inside chasmothecia); relative infection rate; and progress of latency period, for ascosporic *Erysiphe necator* infection on grapevine. The symbol ∇ shows the time of budbreak of vines.

seasonal ascospores that cause infection (i.e. COL) in each infection period, or 'relative infection severity' (Fig. 3c); (vi) progress of latency period (i.e. LATR cumulated starting from any infection period) (Fig. 3c); and (vii) time when secondary infections may begin, i.e. end of the latency period (Fig. 3c). The model generates as many predictions of infection as the number of ascospores discharge events that occur between budbreak and the time when the pool of ascospores is empty (i.e. AIC = 0 and PAR < 1).

Model validation

The model was validated using data from 26 natural epidemics recorded in five grape-growing regions of Italy, representing a 4-year period (2005–2008) with four to 11 vineyards each year (Table 2). Vineyards were selected to be representative of each grape-growing regions in terms of soil types, varieties, training systems and cropping regimes, including fungicides applied in the previous season to control powdery mildew.

None of the 26 vineyards was treated with fungicides against *E. necator* in spring until the time of first disease onset (i.e. until the study was completed). Vines were carefully inspected at least once per week starting from budbreak to determine: (i) the prevalent growth stage of vines (i.e. the growth stage reached by >50% of shoots

in the vineyard) based on the modified Eichhorn-Lorenz system (Coombe, 1995); and (ii) the time of appearance of the first disease symptoms, such as flag shoots or discrete pale spots on the undersides of leaves. For detection of disease signs on leaves, the abaxial surface of leaves positioned close to exfoliating bark of heads and cordons on at least 300 shoots were examined. A sample of these affected leaves was carried to the laboratory to confirm the presence of the disease by microscopic observation. After first detection of the disease, vineyards were sprayed with fungicides against powdery mildew and observations stopped.

Hourly meteorological data of temperature (T, °C), relative humidity (RH, %), rainfall (R, mm), and WD (duration of wetness, h) were supplied by the agro-meteorological regional network for the nearest station (max. 4 km) in Piedmont, Marche and Basilicata, and for the geographic grids (5 × 5 km wide) that comprised the vineyards in Emilia-Romagna (Bottarelli & Zinoni, 2002); a station (Davis Instruments Corp.) was placed within the vineyard of Canneto Pavese.

Weather data were used as model inputs to predict for each vineyard the occurrence of infection periods, their relative severity and the corresponding time of predicted disease onset between budbreak and the time when first disease symptoms actually appeared in the vineyard. The end of latency calculated by the model was used as the

Table 2 Characteristics of the vineyards used for validation of model of ascospore *Erysiphe necator* infection on grapevine

Year	Location	Region	Vineyard-year acronym	Altitude (m a.s.l.)	Grapevine variety	Trellis system	Date of disease onset
2005	Canneto Pavese	Lombardy	CAN05	230	Barbera	Guyot (one arm)	11–18 June
	Bagnacavallo	Emilia-Romagna	BAG05	10	Trebbiano Romagnolo	GDC ^a	22–25 May
	Campremoldo	Emilia-Romagna	CAM05	90	Barbera	Guyot (two arms)	11–18 June
	Castel Bolognese	Emilia-Romagna	CAS05	125	Sangiovese	Guyot (one arm)	20–27 May
	Cotignola	Emilia-Romagna	COT05	18	Trebbiano Romagnolo	GDC	18–25 May
	Bosco Galdo	Basilicata	BOS05	598	Cabernet Sauvignon	Guyot (one arm)	20–27 June
2006	Castell'Arquato	Emilia-Romagna	CAQ06	230	Barbera	Guyot (one arm)	23–30 June
	Rastignano	Emilia-Romagna	RAS06	106	Sangiovese	Guyot (two arms)	10–17 May
	Maltignano	Marche	MAL06	271	Cabernet Sauvignon	Guyot (one arm)	31 May–7 June
	Serrungarina	Marche	SER06	186	Chardonnay	Guyot (one arm)	19–26 June
	Venosa	Basilicata	VEN06	493	Aglianico del Vulture	Guyot (one arm)	15–22 May
2007	Canneto Pavese	Lombardy	CAN07	230	Barbera	Guyot (one arm)	1–8 May
	Castel San Pietro	Emilia-Romagna	CSP07	60	Sangiovese	Spurred cordon	1–8 June
	Castell'Arquato	Emilia-Romagna	CAQ07	230	Barbera	Guyot (one arm)	7–14 May
	Tebano	Emilia-Romagna	TEB07	177	Pinot noir	Spurred cordon	7–14 June
2008	Neive	Piedmont	NEI08	272	Barbera	Guyot (one arm)	27 May–3 June
	Grugliasco	Piedmont	GRU08	285	Moscato	Guyot (one arm)	12–19 May
	Castel San Pietro	Emilia-Romagna	CSP08	60	Sangiovese	Spurred cordon	2–9 May
	Sarna	Emilia-Romagna	SAR08	83	Trebbiano Romagnolo	GDC	21–28 May
	Scipione	Emilia-Romagna	SCI08	250	Sangiovese	Spurred cordon	20–27 May
	Tebano	Emilia-Romagna	TEB08	177	Pinot noir	Spurred cordon	21–28 May
	Agugliano	Marche	AGU08	176	Merlot	Guyot (one arm)	22–29 May
	Maltignano	Marche	MAL08	271	Cabernet Sauvignon	Guyot (one arm)	15–22 May
	Matelica	Marche	MAT08	366	Verdicchio	Guyot (one arm)	20–27 May
	Montefano	Marche	MON08	237	Merlot	Guyot (one arm)	21–28 May
	Serrungarina	Marche	SER08	186	Chardonnay	Guyot (one arm)	21–28 May

^aGeneva double curtain.

estimated time of disease onset on leaves because microscopic examination of these leaves always showed rare fungal colonies bearing conidia. The chosen threshold for predicting an infection period was: relative infection severity >0 .

Predicted disease onset (P) was then compared with actual disease onset (O); all the possible combinations of P vs. O were organized in a 2×2 contingency table, with the following categories: (i) true positives, when the model predicted an infection period and onset of disease that corresponded to actual disease onset in the vineyard (P+, O+); (ii) false positives, when the model predicted an infection period and onset of disease but the disease did not occur (P+, O-); (iii) false negatives, when the model did not predict an infection that actually occurred (P-, O+); and (iv) true negatives, when the model predicted no infection and no disease symptoms appeared in the vineyard during the relevant time period (P-, O-). For determination of the P-, O- category, all the rain events with >0.2 mm of rainfall were considered capable of triggering ascospore discharge and initiating an infection cycle because mature chasmothecia are induced to eject ascospores in the presence of water (Gadoury & Pearson, 1990b; Jailloux *et al.*, 1998; Rossi *et al.*, 2010).

Sensitivity, specificity and accuracy of the model predictions were evaluated by means of the Bayesian analysis (Yuen & Hughes, 2002). The advantages of using the model were assessed by calculating the probabilities that an infection period in a particular vineyard did or did not result in a powdery mildew infection. These probabilities were determined as $P(O+, P+)$ and $P(O-, P-)$ following Madden *et al.* (2008), and were compared with the correspondent prior probabilities, $P(O+)$ and $P(O-)$.

Results

The first seasonal signs of powdery mildew were always observed as discrete small spots on the abaxial surfaces of the basal leaves of shoots; frequently, small veins in the centre of these spots were brown with no visible fungal colonies and a pale yellowish spot was present on the upper leaf surface. Microscopic observations of these spots always showed the presence of a few, superficial hyphae with a small number of conidiophores and conidia typical of *E. necator*. Flag shoots were never found in the vineyard surveys.

The date of first seasonal onset of powdery mildew symptoms in the 26 vineyards ranged from early May to late June (Table 2). In 22 vineyards, disease onset was correctly predicted by the model, even though in several cases the model provided earlier predictions of infection periods that did not result in actual disease onset. At BAG05 (Fig. 4a), for instance, the first predicted infection of the season occurred on 10 May, 15 days after budbreak, when plants were at GS10 (three to four leaves unfolded), with a relative infection severity of 0.13; less severe infection periods were predicted on 13, 16 and 17 May. These predicted infections should have produced symptoms between 20 and 25 May; disease symptoms

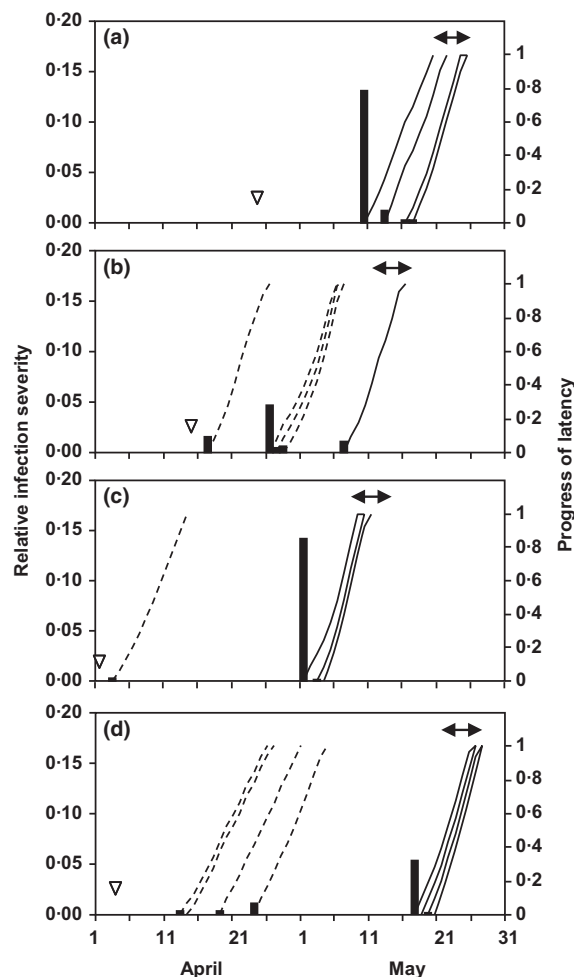


Figure 4 Examples of model output of ascospore *Erysiphe necator* infection on grapevine for four vineyards: BAG05 (a), RAS06 (b), CAQ07 (c) and TEB08 (d). Bars show relative infection rate; lines show progress of latency period for false positive (---) and true positive (—) predictions; the symbol ▽ shows time of budbreak of vines; ↔ observed time of powdery mildew onset in the vineyard. Characteristics of vineyards are reported in Table 2.

were not present on 19 May, but were present on 25 May. At RAS06 (Fig. 4b), the first predicted infection period occurred soon after budbreak (which was on 15 April), and other infection periods were predicted between 26 and 28 April, at GS8 to GS9 (two to three leaves unfolded); none of these predicted infection periods resulted in actual disease onset. A predicted infection period (0.011 relative severity) on 7 May (GS12, five leaves unfolded) should have produced visible disease signs on 16 May; the disease was actually observed on 17 May. Also at CAQ07 (Fig. 4c) the first predicted infection occurred on the 3rd day after budbreak but was not associated with actual disease onset. Later, three infection periods were predicted between 1 and 4 May, when vines were at GS14 (seven leaves unfolded), with maximum relative severity of 0.142; these infections should have resulted in visible symptoms between 10 and 11 May, but

the disease was first observed on 14 May. At TEB08 (Fig. 4d), four infection periods predicted between 13 and 24 April (GS6 to GS8) did not result in disease appearance on leaves, whilst repeated infection periods on 18–22 May (max infection severity = 0.053; GS15, eight leaves unfolded), which should have produced symptoms starting from 27 May, actually resulted in disease onset, which was first detected on 28 May.

In four vineyards (CAM05, MAL06, SER06 and GRU08), the time of disease onset was incorrectly predicted by the model. At CAM05 (Fig. 5), several infection periods with relative severity <0.005 were predicted between 19 April (GS5) and 21 May (GS16, 10 leaves unfolded), but no powdery mildew symptoms appeared in the vineyard until mid-June (Fig. 5b). After 21 May the weather was dry, with no rainfall until 12 June (19.4 mm rain) (Fig. 5a); at that time (GS24, 70% caps off), the model predicted the last infection period of the season (relative infection severity = 0.02) that should have produced symptoms on 20 June, 2 days after the date when disease symptoms were recorded. A similar situation occurred at MAL06 and GRU08 (data not shown). At MAL06, after a 16-day dry period (14–30 May), 34.6 mm of rainfall on 31 May (GS19, beginning of flowering; first flower caps loosening) triggered the last infection period of the season (relative infection severity = 0.007), which should have produced disease symptoms on 11 June; however, the first signs of powdery mildew were observed on 7 June. At GRU08, only light rain occurred between 1 and 12 May (max. 0.8 mm rain); on 13 May (GS12, five leaves unfolded), an infection period initiated by a 2.4-mm rainfall was predicted (relative infection severity = 0.016) and should have resulted

in disease symptoms on 23 May, 3 days after the actual disease appeared on leaves. At SER06, the first powdery mildew symptoms were observed on 26 June, when vines were at GS26 (cap-fall complete); the last seasonal infection period was predicted by the model on 29 May, and no rain fell after 6 June (12.8 mm rain) (data not shown). In the 2 weeks preceding disease onset, the weather was hot (average $T = 25.6^{\circ}\text{C}$) and dry ($\text{VPD} = 17.8 \text{ hPa}$, with no hours of wetness).

Across all 26 vineyards, the model considered 586 rain events between budbreak and appearance of first disease symptoms (or, at SER06, between budbreak and the end of ascospore supply calculated by the model); 34% of these rains resulted in predicted infection and 10% resulted in actual disease onset (Table 3). Accuracy, sensitivity and specificity of the model predictions are shown in Table 3; because of a high proportion of false positive predictions ($\text{FPP} = 0.26$) the model was much better at predicting non-infection periods than infections.

Reasons for high FPP were analysed based on: (i) value of relative infection severity; and (ii) growth stage of vines at the time when any infection period was predicted by the model. Concerning relative infection severity, increasing the threshold for predicting an infection period from 0.0 to 0.15 (i.e. the full range of observed relative infection severity) reduced FPP but also reduced the overall performance of the model. For instance, when the threshold was 0.03 for relative infection severity, FPP was <0.02 but TPP dropped to 0.154 and consequently FNP ($1 - \text{TPP}$) increased to 0.846 (Fig. 6).

Concerning the growth stages of vines, >50% of the predicted infection periods occurred before GS10 (three leaves unfolded), about 40% occurred during shoot growth (between GS11 and GS16, four to 10 leaves unfolded) and <10% occurred after the beginning of flowering (GS17, inflorescence well developed, single flowers unfolded) (data not shown). Frequency of predicted infection periods that actually resulted in disease onset increased as the growth stage advanced (Fig. 7). From budbreak to GS6 (first leaf not completely unfolded from shoot tip), none of the infection periods predicted by the model resulted in infection; 9% and about 56% of predictions resulted in disease onset at GS7–8 (one to two leaves unfolded) and GS11–12 (four to five leaves unfolded), respectively. From GS17 (12 leaves unfolded; inflorescence well developed, single flowers unfolded), 90% of predicted infection periods were correct. Based on these data, an equation was developed that described the probability that a predicted infection period resulted in disease onset as a function of the growth stage of vines at the time of prediction, in the form $P = 1/(1 + e^{-Z})$, where Z is the linear combination: $Z = -5.366 \pm 0.833 + 0.440 \pm 0.068 \bullet \text{GS}$ (Fig. 8); significance ($P < 0.0001$), low standard errors of equation parameters, and values of deviance (=123) and R^2 of Nagelkerke (=0.58) indicated a good fit of the data. Based on this function, the probability that a predicted infection period actually results in disease onset increases as the growth of vines advances; when GS

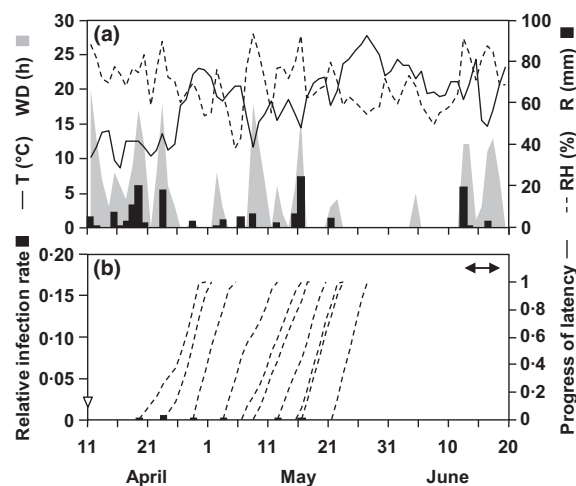
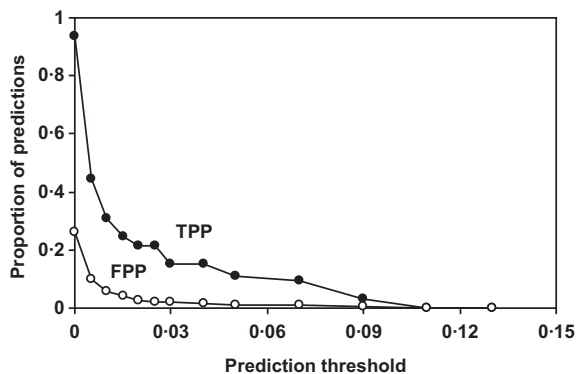


Figure 5 (a) Examples of model input (air temperature, T ; wetness duration, WD ; relative humidity, RH ; and rainfall, R); (b) correspondent output for the vineyard CAM05, for ascosporic *Erysiphe necator* infection on grapevine. In (b), bars show relative infection rate; lines show the progress of latency period for false positive predictions (---); symbol ∇ shows time of budbreak of vines; $\longleftrightarrow$ observed time of powdery mildew onset in the vineyard. Characteristics of the vineyard are reported in Table 2.

Table 3 Comparison between the onset of powdery mildew (*Erysiphe necator*) predicted by the model and the first signs of disease observed in 26 vineyards, in 2005–2008

Observed	Predicted		Total	Accuracy ^a	Likelihood ratio (<i>LR</i>)	Prior probability (<i>P</i>)	Posterior probability (<i>P</i>)	
	Yes (<i>P</i> +)	No (<i>P</i> -)						
Yes (<i>O</i> +)	61 (0.94) ^b	4 (0.06) ^c	65	0.76	<i>LR</i> (<i>O</i> +)3.6	<i>P</i> (<i>O</i> +) 0.11	<i>P</i> (<i>O</i> +, <i>P</i> +)0.31	<i>P</i> (<i>O</i> +, <i>P</i> -) 0.01
No (<i>O</i> -)	136 (0.26) ^d	385 (0.74) ^e	521		<i>LR</i> (<i>O</i> -)0.08	<i>P</i> (<i>O</i> -) 0.89	<i>P</i> (<i>O</i> -, <i>P</i> +)0.69	<i>P</i> (<i>O</i> -, <i>P</i> -) 0.99
Total	197	389	586					

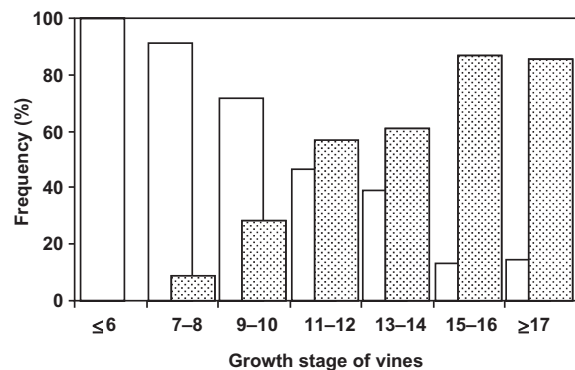
^a(*P*+,*O*+/*P*-,*O*-)/total cases.^bTrue positive proportion, TPP (sensitivity).^cFalse negative proportion, FNP (set to 0.001 for calculation of *LR* and *P* for 2006 and 2007).^dFalse positive proportion, FPP.^eTrue negative proportion, TNP (specificity).**Figure 6** Proportions of true positive (TPP) and false positive (FPP) infection periods predicted by the model of ascospore *Erysiphe necator* infection on grapevine, as affected by the use of different prediction thresholds for the relative infection rate.

increases by 1, the odds increase by a factor of 1.55 (the odds are defined as the ratio of the probability that the event will occur to the probability that it will not). For instance, the probability is $P < 0.5$ at growth stage $\leq \text{GS}12$ (five leaves unfolded) and $P > 0.9$ at growth stage $\geq \text{GS}18$ (14 leaves unfolded; flower caps still in place, but cap colour fading from green).

Discussion

This research used a mechanistic approach to develop a new model for *E. necator* ascospore infections. Mechanistic modelling uses fundamental knowledge of the interaction between variables in the system to define the model structure, and requires the precise definition of functional relationships between these variables (Rossi *et al.*, 2010). Mechanistic models can increase the understanding of pathosystem behaviour and can lead to prediction of disease under a range of circumstances (Campbell & Madden, 1990).

The model was evaluated over a 4-year period in five vine-growing areas of Italy by comparing infection periods predicted by the model with actual observations of the first seasonal symptoms of powdery mildew in 26 vineyards. The model showed high sensitivity, specificity

**Figure 7** Frequency of true positive (shaded bars) and false positive (white bars) proportions of the infection periods predicted by the model of ascospore *Erysiphe necator* infection on grapevine versus the growth stage of vines at the time of prediction: ≤ 6 (budbreak to first leaf not completely unfolded from shoot tip), 7–8 (one or two leaves unfolded), 9–10 (two or three leaves unfolded), 11–12 (four or five leaves unfolded), 13–14 (six or seven leaves unfolded), 15–16 (eight to 10 leaves unfolded), ≥ 17 (inflorescence fully developed, single flowers unfolded).

and overall accuracy (Madden *et al.*, 2008). Moreover, the likelihood ratios of infection and no infection periods showed good accuracy because an accurate model generally has large *LR* (*O*+) (>1) and small *LR* (*O*-) (close to 0) (Madden *et al.*, 2008).

Because of the properties of the model and the prior probability of an infection, users of the model would have more confidence in prediction of non-infection periods than in prediction of infections. In fact, the achieved probability that a predicted non-infection period did not produce infection was 0.99, whilst the prior probability was 0.89; however, there was still a small probability (0.01) that disease symptoms occurred when disease had not been predicted by the model. These incorrect negative predictions occurred in four vineyards. In three of them (CAM05, MAL06 and GRU08), an infection period was predicted that should have resulted in disease onset 2–4 days later than the date when the first disease signs were observed on leaves. Because the environmental conditions registered in these three vineyards were unsuitable

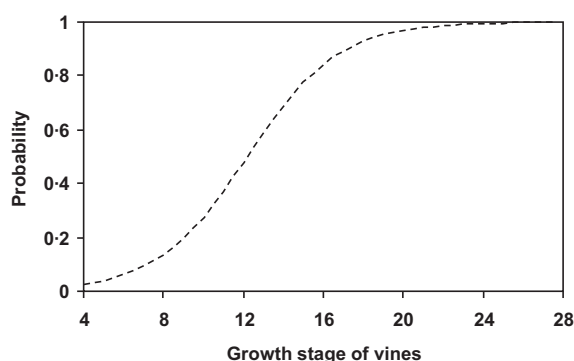


Figure 8 Probability distribution that a predicted infection period actually results in powdery mildew (*Erysiphe necator*) onset as a function of the growth stage of grapevines at the time of prediction. Probability distribution was estimated by a logistic regression in the form $P = 1/(1 + e^{-Z})$, where Z is the linear combination: $Z = -5.366 + 0.440 \text{ GS}$, where GS is the growth stage according to the modified Eichhorn-Lorenz system (Coombe, 1995).

for *E. necator* infection for long time before actual disease onset, the error probably resulted from an overestimation of the latency period. However, this error would not have had a practical impact on disease control because a fungicide applied at the time when the model predicted the infection period (i.e. 6–7 days before observed disease onset) would have provided efficient control. In the fourth vineyard (SER06), the first disease symptoms recorded in late June were preceded by a long period with hot and dry conditions, more suitable for conidial (Delp, 1954) than for ascospore (Gadoury & Pearson, 1990a) infection; therefore, it is possible that the observed symptoms were caused by conidia that originated from outside or from a previous, undetected infection caused by ascospores.

Model performance in predicting a true infection period was characterized by a considerable false positive proportion (FPP = 0.26). For a powdery mildew infection to occur, it is necessary that some ascospores are ejected from chasmothecia, that some of them are deposited on a susceptible host surface, and that they experience favourable environmental conditions. The present model accounts for the first and the third stages of this process and then estimates the relative infection severity of any predicted infection period, i.e. the proportion of the seasonal ascospore dose which is able to cause infection. Model evaluation showed that there was not a close relationship between calculated values of relative infection severity and actual infection: true positive predictions occurred with either very low (min. 0.001) or high (max. 0.13) severity values. What is lacking in this model is an estimate of the actual inoculum reaching leaves, which depends on: (i) the amount of overwintered inoculum in a vineyard (the initial inoculum dose), and (ii) the probability that the part of this inoculum that becomes airborne encounters a leaf; this probability depends, in turn, on extension of the leaf area and hence growth stage at the time of ascospore ejection.

The initial inoculum dose of the vineyards used for model validation was not estimated; however, numbers of chasmothecia that are dispersed to vine trunks in a vineyard can range from <0.1 to $>1.5 \text{ cm}^{-2}$ (Rossi *et al.*, 2009). The concept of 'clean' and 'dirty' vineyards developed by Magarey *et al.* (2000) strongly supports the relevance of the overwintering inoculum dose in determining infection severity in the following spring: in clean vineyards, the low levels of chasmothecia ready to initiate disease make it possible to use less intensive spray programmes early in the season (Emmett *et al.*, 1997). Growth stage of vines was determined in each vineyard at the time of any predicted infection period; most of the wrong predictions of infection occurred when vines were at early growth stages, with fewer than three or four leaves unfolded. In these stages, the extent of the host surface susceptible to infection was very low and so the probability that ascospores ejected into the air encountered susceptible host tissue was also very low, such that a predicted infection may not have resulted in disease. In another model for disease spread in a vineyard (Calonnec *et al.*, 2008), the quantity of powdery mildew conidia deposited on a leaf depends upon the leaf's distance from the spore source and was proportional to its surface area. It follows that the current model can be improved to reduce false positive predictions; these improvements involve the addition of: (i) an estimate of the inoculum dose of each vineyard, and (ii) a probability function relating predicted infections to canopy size or growth stage of vines. Although the present work includes an equation describing the probability that a predicted infection period result in disease onset as a function of growth stage of vines at the time of prediction, this equation must be validated against independent data.

In conclusion, the model presented in this work represents a considerable enhancement in modelling the sexual stages of *E. necator* that lead to primary infections. Considering that (i) the dataset used for validation was not used in model building; (ii) the dataset was representative of different epidemiological conditions, with both early and late *E. necator* infections; and (iii) neither calibration nor empirical adjustment of model parameters was necessary to obtain accurate prediction, one could state that this model produces a reasonable approximation of the ascospore infection cycle of *E. necator*.

The new model meets the need for a forecasting system that predicts epidemic initiation by ascospores (Calonnec *et al.*, 2006). Fungicide sprays could be avoided when the model does not predict an infection period and initiated only when the model predicts an infection. In this case, a certain probability of unjustified sprays persists, but this represents an improvement because it will produce fewer unjustified early season treatments than the current spray schedules. These unjustified sprays can be reduced by using model predictions of infection to schedule the scouting of the vineyard for detection of first disease signs (Carisse *et al.*, 2009); this would reduce labour costs because early detection of the disease in the vineyard is

difficult and requires expert and intensive sampling, which is time consuming.

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